

PRIYANSHU PANT

Dehradun, Uttarakhand

+91-9548943635 pantpriyanshu707@gmail.com [in](#) [Linkedin](#) [G](#) [Github](#) [Codeforces](#) [LeetCode](#) [GeeksforGeeks](#)

EDUCATION

Graphic Era Hill University ,Dehradun

2023 – 2027

B.Tech - Computer Science and Engineering - CGPA - 8.70

Dehradun, Uttarakhand

COURSEWORK / SKILLS

- DSA
- OOPS Concepts
- System Design
- Software Engineering
- Operating Systems
- Computer Networks
- Cloud Computing
- DBMS

PROJECTS

LPG Gas Pipeline Optimization System | [FastAPI](#), [React.js](#), [A*](#), [Prim's Algorithm](#)

Jan 2025

- Designed and developed a full-stack pipeline optimization system using A* pathfinding and Prim's MST algorithm.
- Built an interactive grid-based visualization in React to simulate gas station-to-house connections with obstacle handling.
- Improved routing efficiency and reduced computation time by optimizing spatial search logic.
- Created REST APIs using FastAPI to process grid data and return optimized pipeline networks.

Dental Clinic Management System | [MERN Stack](#), [JWT](#), [Role-Based Access](#)

Feb 2025

- A secure clinic management platform supporting multiple user roles including admin, doctor, and patient.
- Applied authentication and authorization using JWT-based access control.
- Built features for appointment scheduling, patient records management, and doctor dashboards.
- Integrated MongoDB database with Express APIs for scalable data handling.

AI Image Colorizer & Cartoonizer | [Python](#), [OpenCV](#), [Deep Learning](#), [React](#)

Mar 2025

- Built an AI-powered image enhancement tool for grayscale colorization and cartoonization.
- Integrated deep learning image processing pipeline with a React-based user interface.
- Added real-time preview, download functionality, and theme switching for improved UX.
- Enforced animation using Framer Motion for smoother user interaction.

AI Assistant (RAG-based LLM System) | [Python](#), [LangChain](#), [FAISS](#), [FastAPI](#), [LLM APIs](#)

Jan 2026

- A Retrieval-Augmented Generation (RAG) system to answer questions from custom documents using LLMs.
- Built an agentic workflow that retrieves context from a vector database and generates grounded responses.
- Implemented document embedding, semantic search, and prompt orchestration using LangChain.
- Created FastAPI endpoints for document upload, indexing, and conversational querying.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL, VBScript

Technologies/Frameworks: HTML5, CSS3, Tailwind CSS, React.js, Node.js, Express, FastAPI, MongoDB, PostgreSQL, SQLite

Developer Tools: VS Code, PyCharm, IntelliJ IDEA, Git, GitHub, Canva

Others: JWT Authentication, REST APIs, Algorithms & DSA

CERTIFICATIONS

- Network Security - Great Learning
- Front End Development - Great Learning
- Google Analytics - Great Learning
- AWS Cloud Practitioner - AWS

ACHIEVEMENTS

- Solved 300+ questions on leetcode
- Selected for Microsoft SEFA program
- Mentored juniors in AI Web Development projects